● ROUND 1

FLEXIPATH – SWITCH BEFORE THE LINK FAILS

Keep a drone's control link, voice and video alive when it moves between Wi-Fi, 5G, satellite and wired networks. Switch before the signal fails.

TEAM

TEAM TAROT CLUB
Al Ain University

EDGE GROUP CHALLENGE
Advanced Technology Pioneers 2026

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LIVE PROTOTYPE
flexipath.vercel.app

▶ TRY THE LIVE PROTOTYPE
flexipath.vercel.app
guided demo, simulator and the 2-minute video

EXECUTIVE SUMMARY

FlexiPath keeps critical connections alive with little extra power

PROBLEM

In our model, a legacy stack leaves one network before it joins the next. A moving user loses the connection. When the control link (C2) of a drone is silent for more than 300 ms, the drone must stop and hover (our piloted-flight rule).

MECHANISM

FlexiPath finds a fading signal seconds early. It wakes the next network and tests it. For 0.4 s, it sends only the critical commands on both networks. Then it switches with no gap. An anchor relay keeps each connection open.

Spare modems sleep at all other times to save power.

EVIDENCE

We run 160 paired simulated flights (8 scenarios, 5 seeds, 4 speeds) with a legacy stack and two existing methods. Reactive multipath (MPTCP/ATSSS) switches after failure. Always duplicate (PDCP duplication) sends twice.

With our 300 ms rule and 60 vs 250 ms outage detection, FlexiPath keeps C2 on 100 % of flights. A legacy stack keeps it on 0 %. Reactive multipath fails in three scenarios. Always duplicate uses 2.1–2.6× the power.

DELIVERED

4–7 points more reliability per extra watt for FlexiPath versus 0.3–0.9 for always duplicate.

A working prototype (interactive 3D digital twin with a guided demo), a deterministic simulation with a headless benchmark, and a documented model. We check its parameters against published data.

All results are from simulation; we claim no live-network measurements.

THE PROBLEM AND WHY IT MATTERS

In our model, every network change costs a legacy stack **3.8 s** of silence.

Dropped calls. Frozen streams.
300 ms of silence makes a drone hover.

Emergency response, industrial automation, connected transport and defence cannot wait.

Two drones, one route: Wi-Fi → 5G → satellite.



Why drones: we model a hover after 300 ms without C2 (our piloted-flight rule).

1. DOCK (WIRED)
Stable baseline mission prep

2. WI-FI 6
Local coverage on the move

3. 5G SA
Wide-area coverage at speed

4. LEO SATELLITE
Global reach
beyond line of sight

CONTROL LINK (C2)
Command and control

PUSH-TO-TALK VOICE (REAL-TIME)
Clear, low-latency communications

CAMERA UPLOADS (BULK)
High-throughput data transfer

EXISTING MECHANISMS WE ANALYSED (CRITERION 01)

REACTIVE MULTIPATH

MPTCP backup subflow (RFC 8684) and 3GPP ATSSS active-standby.

Keeps a backup network on standby. It moves only after it confirms losses and the spare modem wakes up.

0.35 s gap, more than our 300 ms C2 budget

1.20 W low power consumption

ALWAYS DUPLICATE

MPTCP redundant scheduler and PDCP duplication.

Sends every important packet on two networks, all the time.

Still fails when both paths fail together (a shared radio shadow).

99–100 % critical delivery (most reliable)

2.65–3.39 W high power consumption and copies 49–100 % of protected bytes

COMMON SHORTCOMINGS

- ✕ Neither looks ahead.
- ✕ Ignores the fading signal.
- ✕ Ignores the known route.

Neither adds protection only at the moment a link fades.



WHAT WE ADAPT

Keep the **anchored session** (from multipath).
Keep the **duplication** (from PDCP), but use it only in a **400 ms** burst around a predicted failure.
New in combination: **prediction**, **route knowledge** and **traffic classes**.

References: RFC 8684 MPTCP v1 (2020) · 3GPP TS 23.501 §5.32, ATSSS · 3GPP TS 38.323, PDCP duplication · 3GPP TS 22.125 §7.2, UAV C2 KPIs · Related: Elsight Halo, Peplink SpeedFusion, uAvionix muLTElink

THE FLEXIPATH MECHANISM (CRITERION 02)

01 DETECT

Measure the primary link ten times a second.

Spare modems sleep in DRX at 1.2 W and report signal once per cycle.

02 PREDICT

Extend the signal trend to find where packet loss reaches 15 %.

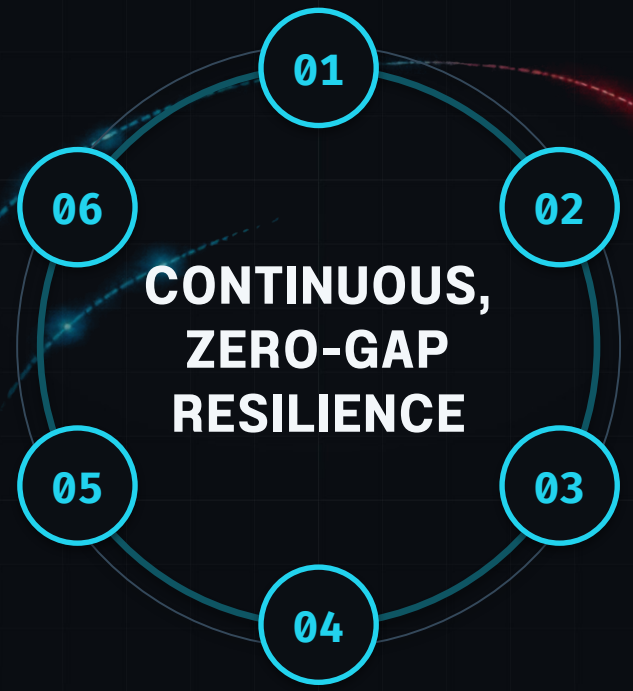
Watch from 8 s to failure; act under 3 s.

The coverage map warns 8 s before a known shadow.

03 PREPARE

Wake the best spare modem (1.8 W) and test it with small 64-byte heartbeats.

Check its delay and loss before it carries anything.



04 REPLICATE

For a 400 ms burst (3.4 W), send critical traffic on both paths. Add real-time traffic if its loss is over budget.

The receiver keeps the first copy of each sequence number and discards the late duplicate, so head-of-line delay is 0 ms.

Never copy bulk traffic; only pace it.

05 HONEST BEHAVIOUR

A reversed trend cancels the prediction at no cost.

If an outage comes with no warning, use a fast failover (160 ms in our model).

06 SWITCH AND CONTINUE

Make the tested network the main one with a 0 ms gap. Put the old radio back to sleep.

Connections stay open in a tunnel to the anchor relay.

SYSTEM DESIGN

One engine. Multiple radios. Continuous sessions.

ON VEHICLE

APPLICATIONS

- C2 link (critical)
- push-to-talk voice (real-time)
- camera and survey **uploads (bulk)**

FLEXIPATH ENGINE ON VEHICLE

- link monitor and predictor
- path scorer
- **traffic manager** replication, pacing, reordering buffer
- **radio power manager** DRX / warm probe / burst
- coverage map

RADIOS

- wired dock port
- Wi-Fi 6
- 5G SA
- LEO satellite

link model: signal, loss, RTT, jitter, capacity

ANCHOR RELAY

a fixed address in the operator's network

the vehicle's tunnel rebinds across radios

the remote side never changes

EXPLAINABILITY AND DASHBOARD

it records each decision with the values at that tick:

current path | trend | time-to-failure | alternate score | radio tier | action | reason

DESIGNED TO SIT UNDER ANY COMMAND-AND-CONTROL APPLICATION

it needs no change to the application

PROTOTYPE, SIMULATION AND MODEL (CRITERION 03)

flexipath.vercel.app

interactive 3D digital twin with mission and advanced views

GUIDED DEMO

pauses at each real decision, explains what happened and which panel shows it

DETERMINISTIC 10 HZ ENGINE

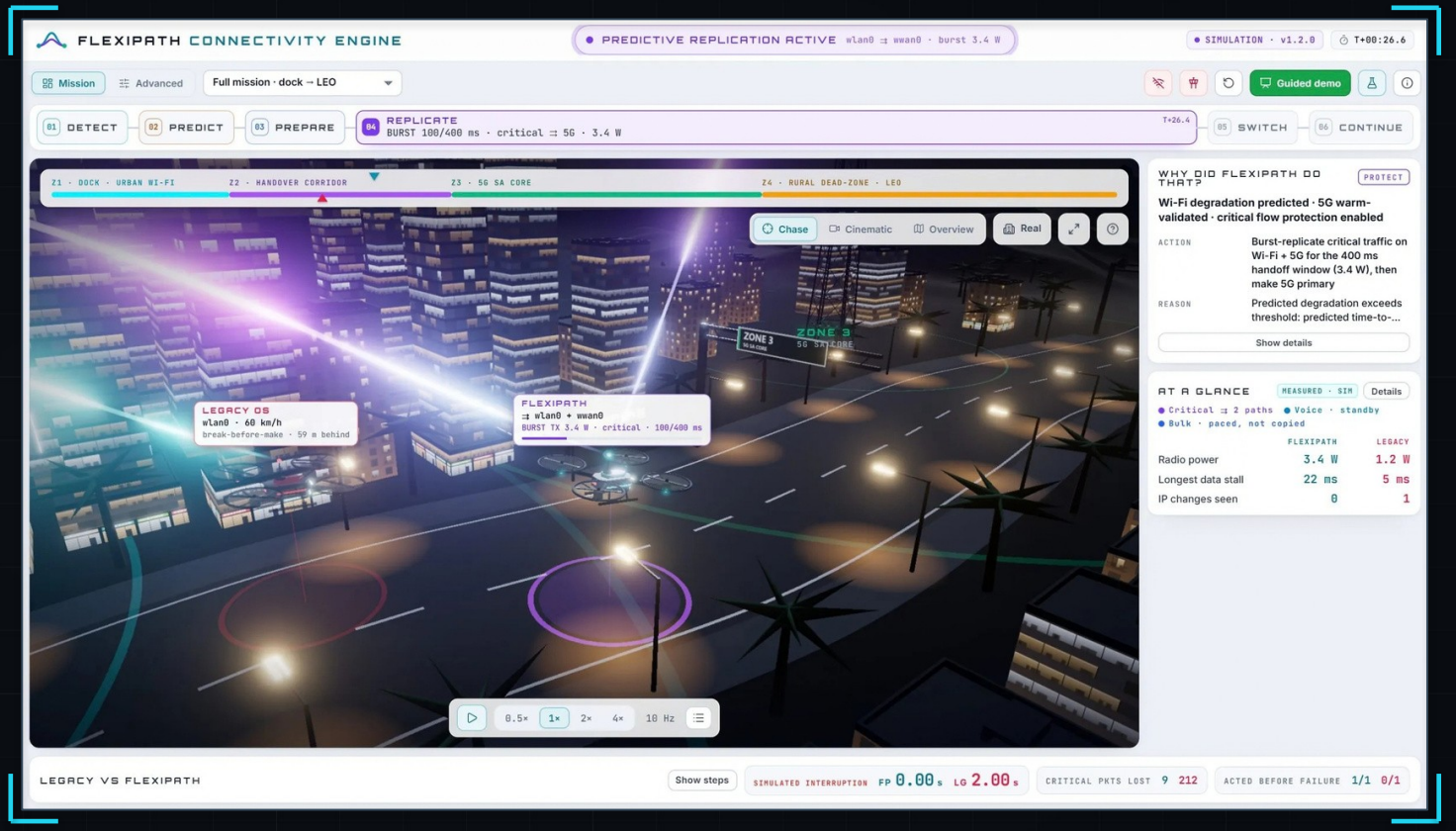
seeded random streams with fading, capacity, loss, RTT, jitter, TCP and radio models

HEADLESS BENCHMARK

scenarios A–H × 5 seeds × 4 speeds = 160 paired runs per stack, plus baselines and ablations; CSV output with 95 % CIs

PUBLISHED MODEL

we check parameters against published measurements and test the 100 ms modem wake at 260 ms as a sensitivity run



Mission view of the live prototype. The FlexiPath drone (cyan) and the legacy-stack drone (crimson) fly with the same seed and radio conditions. The decision loop, decision card and scoreboard are around the twin.

▶ **Try it yourself:** flexipath.vercel.app ·
the guided demo, the simulator and the 2-minute video

RESULTS AGAINST THE EXISTING MECHANISMS (CRITERION 04)

Paired means over 20 runs per scenario, same seeds for every stack.

01

100%

of flights where FlexiPath keeps C2 (flight count) under our piloted 300 ms rule

02

1.2–4.6 s

FlexiPath acts before the path fails, versus 0.1–0.5 s after for reactive multipath

03

35%

fewer critical packets lost vs reactive multipath (CI 30–39 %); held-out; ~20 % with equal detection

04

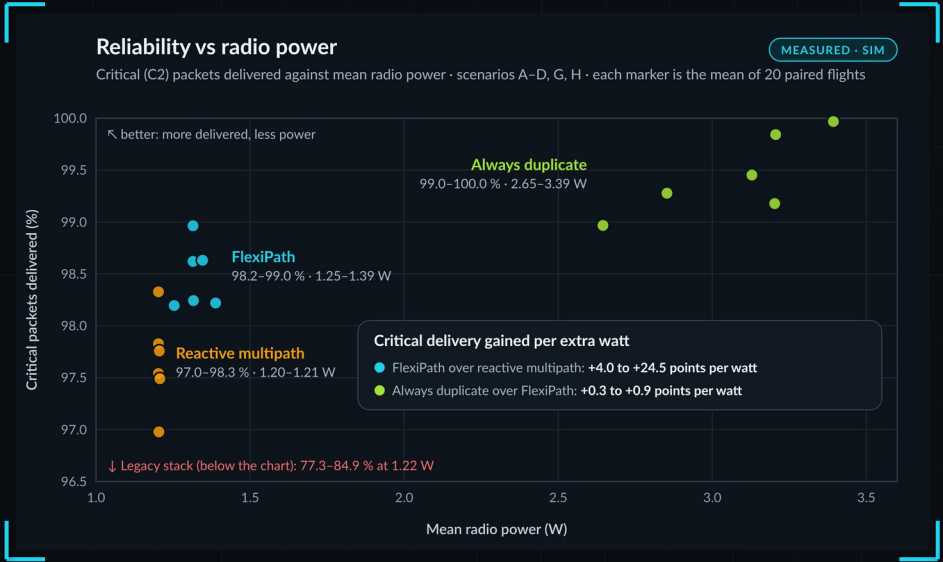
4.0–7.1 pp/W

Reliability per extra watt versus reactive multipath **(16.9–24.5 in D)**

05

4.5–21×

FlexiPath's extra watts give 4.5–21× more reliability than always duplicate's **0.3–0.9 pp/W**



Reliability against radio power, scenarios A–D, G and H (paired means). FlexiPath is between the two existing designs. It gives 0.6–1.2 points more critical delivery than reactive multipath for 0.05–0.18 W more. It uses 2.1–2.6× less power than always duplicate.

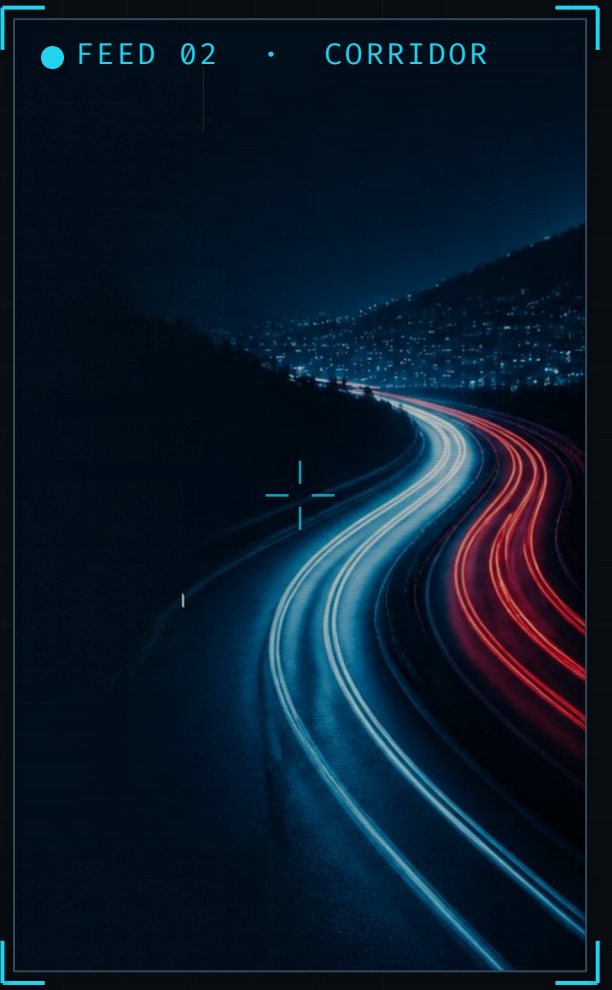
Flights that keep C2 under the piloted 300 ms rule

	Legacy	Reactive multipath	Always duplicate	FlexiPath
A, B, C	0	100	100	100
D	0	0	100	100
G, shared radio shadow	0	0	40	100
H, map out of date	0	0	100	100

Robustness: the ablations show trend prediction gives the gain; in our model, the burst adds no measurable reliability. The results hold on held-out seeds. Under the 1 s waypoint rule, every multipath stack keeps C2. So the piloted-rule advantage needs a spare modem that resumes fast.

GUIDELINES FOR QOS AND QOE ACROSS HETEROGENEOUS NETWORKS (CRITERION 05)

01	Anchor the session, not the interface — a fixed **anchor relay** prevents every reset.
02	Act on the trend, not on the failure — a **3 s signal trend** gives seconds of warning.
03	Prepare the next path just in time, and only then — test it at **1.8 W**; do not keep radios on.
04	Duplicate only what needs it, only around the handover — a **400 ms burst** costs ≈ 1 % of protected bytes.
05	Deliver the **first copy**; never make real-time traffic wait in order.
06	Detect a failed path faster than the application allows — within the **300 ms C2** budget.
07	**Hold still** unless a move clearly helps — a reversed trend cancels the prediction at no cost.
08	Measure what **users feel**, on matched pairs: interruption, delivery, voice MOS, power, same seeds.
09	Use what you know about the route — a **coverage map** is better than reaction if two radios fail together.



PLAN FOR REAL-LIFE USE, LIMITS, AND DECLARATION

Deployment path, known limits, and our AI-use declaration

DEPLOYMENT

A userspace agent on the vehicle's companion computer (Linux) controls modems through standard interfaces.
The anchor relay is at the operator's ground station or cloud edge.
It runs under current C2 and video applications; they need no change.

LIMITS

All results are simulated.
We evaluate the radio model at ground level.
Piloted rule needs a spare modem that resumes in ~100 ms (5G RRC_INACTIVE or connected-mode DRX).
At 260 ms, an unexpected outage still forces a hover.

REAL-LIFE USE

Phase 1 (bench): laptop with Wi-Fi and a 5G dongle runs the included handover probe to measure one real network change. Phase 2 (field): test drone with Wi-Fi, 5G and a LEO terminal on a surveyed corridor; first survey flights build the coverage map. Phase 3 (pilot): one operator, one corridor, the paired-comparison method of this study applied to real flights.

AI-USE DECLARATION

AI tools (Anthropic Claude) helped with code, analysis drafts, this deck and the video. An AI voice (Kokoro) reads the narration. The design decisions, the experiments and their interpretation are the team's own, and every member can explain their part in their own words.

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